The Complete Roadmap to Wireless and Mobile Network

Delivery Plan (Weekly Syllabus) - Theory

Week	Official Module Title
Week 1	**Fundamentals of Wireless Transmission & Architectural Differences**.
Week 2	**Cellular Evolution & 5G/6G Architectures** .
Week 3	**Signal Processing, Modulation, & Multiplexing** .
Week 4	**Medium Access Control (MAC) Layer Principles** .
Week 5	**Coordinated & Spread Spectrum Access** .
Week 6	**Random Access & The Evolution of Wi-Fi (IEEE 802.11)** .
Week 7	**Mid-term Exam Week**
Week 8	**Mobile Network Layer Issues & Mobility Management** .
Week 9	**Software-Defined Wireless Routing & Ad-hoc Networking** .
Week 10	**Transport Layer Issues in Mobile Environments** .
Week 11	**Next-Generation Transport Protocols: From Wireless TCP to QUIC**.
Week 12	**Mobile Network Performance Optimization** .
Week 13	**Mobile Edge Computing (MEC) & Cloud-Native Data Handling** .
Week 14	**Wireless Localization, Modeling, and Simulation** .
Week 15	**Preparatory Week & Case Studies**.

Table 1: Module Delivery Framework

Copyright © 2026 Dalya Raad AlAni

Weekly Theory Blueprint (15 Weeks)

1. **Week 1: Fundamentals of Wireless Transmission & Architectural Differences**
   (a) Reviewing key structural and physical properties differentiating wired networks from wireless environments. Foundational overview of electromagnetic wave propagation, signal attenuation, free-space path loss, multipath fading, and basic antennas.
2. **Week 2: Cellular Evolution & 5G/6G Architectures**
   (a) Tracking the architectural leap from 1G–4G to cloud-native 5G Standalone (SA) and Non-Standalone (NSA) frameworks. Introduction to foundational next-generation structures: User Plane Functions (UPF), Access and Mobility Management Functions (AMF), Network Slicing, and early 6G design paradigms.
3. **Week 3: Signal Processing, Modulation, & Multiplexing**
   (a) Analysis of physical layer efficiencies. Concepts of multiplexing (FDM, TDM, CDM) and digital modulation schemes. In-depth focus on Orthogonal Frequency Division Multiplexing (OFDM) and Massive MIMO configurations powering modern wireless standards.
4. **Week 4: Medium Access Control (MAC) Layer Principles**
   (a) The role of the MAC layer in wireless networks. Managing collisions, hidden and exposed terminal issues, and near-far interference. Analysis of contention-based vs. contention-free access mechanics.
5. **Week 5: Coordinated & Spread Spectrum Access**
   (a) Structural breakdown of structured multiple access techniques including FDMA, TDMA, and CDMA. Study of spread spectrum principles: Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS).
6. **Week 6: Random Access & The Evolution of Wi-Fi (IEEE 802.11)**
   (a) Deep dive into CSMA/CA, RTS/CTS mechanisms, and frame formats. Transitioning from legacy 802.11 standards directly into the structural and efficiency improvements of Wi-Fi 6 (802.11ax) and Wi-Fi 7 (802.11be).
7. **Week 7: Mid-term Exam Week.**
8. **Week 8: Mobile Network Layer Issues & Mobility Management**
   (a) Examining how spatial user mobility breaks traditional routing tables. Addressing handoff mechanisms, location management, and tracking. Transitioning from legacy Mobile IP components to modern session management in cellular core routing.
9. **Week 9: Software-Defined Wireless Routing & Ad-hoc Networking**
   (a) Mobile Ad-hoc Networks (MANETs) and Wireless Sensor Networks (WSNs). Routing challenges without fixed topology (AODV, DSR protocols). Introduction to Software-Defined Wireless Networking (SDWN) for dynamic routing.
10. **Week 10: Transport Layer Issues in Mobile Environments**
    (a) Investigating the conflict between standard TCP congestion control and the physical realities of wireless connections. Analyzing why packet drops from wireless fading are misidentified as network congestion, causing throughput degradation.

Copyright © 2026 Dalya Raad AlAni

11. **Week 11: Next-Generation Transport Protocols: From Wireless TCP to QUIC**

    (a) Brief review of legacy optimizations (Indirect-TCP, Snoop-TCP). Comprehensive transition to QUIC (Quick UDP Internet Connections) / HTTP/3, focusing on connection-migration mechanics during mobile roaming and zero-RTT handshakes.

12. **Week 12: Mobile Network Performance Optimization**

    (a) Defining Quality of Service (QoS) in wireless topologies. Latency, jitter, packet delivery ratio, and throughput modeling. Channel assignment strategies and link adaptation protocols.

13. **Week 13: Mobile Edge Computing (MEC) & Cloud-Native Data Handling**

    (a) Replacing legacy localized data collection structures with Multi-access Edge Computing (MEC) architectures. Understanding how low-latency computing resources are deployed near the cellular network edge to process real-time application pipelines.

14. **Week 14: Wireless Localization, Modeling, and Simulation**

    (a) Spatial localization models (triangulation, fingerprinting, GPS infrastructure). Framework configuration for simulation and modeling using industry-relevant tools like NS-3 or Mininet-WiFi.

15. **Week 15: Preparatory Week & Case Studies**

Copyright © 2026 Dalya Raad AlAni

References

Dordal, P. L. (2023). *An introduction to computer networks* [Release 2.0.11, Open-Access (CC BY-NC-ND), ISBN: N/A]. Loyola University Chicago Department of Computer Science.

Du, K.-L., & Swamy, M. N. S. (2026). *Wireless communication systems*. Springer.

Molisch, A. F. (2022). *Wireless communications: From fundamentals to beyond 5g* (3rd) [ISBN: 978-1-119-11721-6]. Wiley-IEEE Press.

Peterson, L., & Sunay, O. (2020). *5g mobile networks: A systems approach* [ISBN: 9781681738888]. Morgan & Claypool.

Tanenbaum, A. S., & Wetherall, D. J. (2010). *Computer networks* (5th). Pearson.

Copyright © 2026 Dalya Raad AlAni